Quantum Tunnelling

Voiceover script · about 62 seconds

Physics, please.
What is this quantum physics nonsense?
Right now, inside the Sun, particles are doing something impossible.
Roll a ball at a hill without enough energy, and it rolls back.
Physics calls that hill an energy barrier.
No energy, no crossing.
But electrons are described by a wave, and a wave doesn’t stop dead at a wall.
Then, inside the barrier, it fades fast, but if the wall is thin, a little wave makes it through.
That leftover wave is a real chance the particle turns up on the other side.
That’s quantum tunnelling.
Thicker walls and heavier things make the odds crash.
You’re about ten to the twenty-eight atoms.
Stop trying your bedroom wall.
You see, the Sun’s protons repel each other, and even at fifteen million degrees, almost none are fast enough to fuse.
But enough of them tunnel through, every second, to power the Sun.
It also runs your phone’s flash memory, and a microscope that sees single atoms.
So no, it’s not teleporting, and it’s not faster than light.
Just waves and odds.
Every sunny day is a particle doing something it isn’t allowed to do.